For this lab, we were tasked with creating a humanitarian design incorporating a moisture sensor, a photoresistor, and an LED, created with hot glue, popsicle sticks, Knex, and a few other materials. We came up with the idea for an automatic roof, which would close in wet or dark conditions and open in dry and light conditions. Some possible applications for this project could be in a sports stadium or residential patio, increasing the comfort of its users. Initially, we thought of two main prototypes: a swing-arm and a roll-up roof. In prototyping, the roll-up roof appeared to be simpler in its construction than the arm, and thus we decided to use that. As we built our prototype, we discovered that the structure opened just fine, but had issues closing. To remedy this, we added a secondary axle with a string attached, which reliably allowed the roof to close. From there, we integrated a stepper motor with a belt and pulley drive to drive the two axles. Initially, the belt was slipping whenever the knot in the belt was on the drive motor's pulley, but this was solved by rotating the belt, giving the pulley more traction. Once we had gotten the mechanical system working, we began work on the electronics.

The electronics appeared to be straightforward, but raised a few issues. We used an Arduino Nano ESP32, a homemade moisture sensor, a photoresistor, an LED, and a switch to create the system. The wiring proceeded without issue, and we used Gemini to write the code for the project. The one major issue that arose was that our moisture sensor was not outputting a usable reading. After around two hours of troubleshooting, we discovered that we needed to add a larger resistor to the circuit and adjust the code so that an increase in moisture would be measured by the code. After implementing this change, the entire system worked well. The roof opened and closed exactly how we wanted it to based on the correct inputs, the LED flashed while the roof was in motion, and the switch enabled and disabled the system. Overall, the final prototype was a great success.

This project allowed every person in my group to improve our overall engineering and collaboration abilities. We ran into several problems with our system, but through troubleshooting and teamwork we were able to remedy every single one of them, resulting in a project to be proud of. The most difficult part of the process was trying to fix the moisture sensor output, but that was solved eventually, leading to the success of the system. I believe that I am a better engineer than I was before this project and module, largely due to the practice in solving hands-on problems throughout the project.